

Algorithmic Pseudocode Sample 9

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

```
1:  $t \leftarrow 1$ ;  $\mathbf{m}_1 \leftarrow \mathbf{1}_{\bar{C}}$ ;  $\alpha \leftarrow 1$  # Step  $t = 1$ 
2: for  $e \leftarrow 1, \dots, E$  do
3:   FBP() # Algorithm ??
4: end for
5:  $\bar{\mathbf{s}}_1 \leftarrow$  mean of  $\tilde{\mathbf{s}}$  across data
6: for  $t \leftarrow 2, \dots, T$  do # Steps  $t \geq 2$ 
7:   # Phase 1
8:    $\bar{\mathbf{s}}_t \leftarrow \bar{\mathbf{s}}_{t-1}$ ;  $\alpha \leftarrow \frac{1}{2}$ ;  $\mathbf{m}_t \leftarrow \mathbf{m}_{t-1}$ 
9:   for  $e \leftarrow 1, \dots, E_1$  do
10:    FBP()
11:  end for
12:   $\bar{\mathbf{s}} \leftarrow$  mean of  $\tilde{\mathbf{s}}$  on data # Phase 2
13:   $\bar{\mathbf{s}}_t \leftarrow \frac{1}{2}(\bar{\mathbf{s}}_t + \bar{\mathbf{s}})$ 
14:  for  $e \leftarrow E_1 + 1, \dots, E_2$  do
15:     $\alpha \leftarrow \max\{\alpha - \frac{1}{2(E_2 - E_1)}, 0\}$ 
16:    FBP()
17:  end for
18:  # Indices that sort an array; Phase 3
19:   $R = \text{argsort}\{\bar{\mathbf{s}}_t[i] : \mathbf{m}_t[i] = 1\}$ 
20:   $D_t \leftarrow \{R[0], \dots, R[C_{t-1} - C_t]\}$ 
21:  for  $e \leftarrow E_2 + 1, \dots, E_3$  do
22:     $\mathbf{m}_t \leftarrow \max\{\mathbf{m}_t - \frac{\mathbb{I}[i \in D_t]}{E_3 - E_2}, \mathbf{0}_{\bar{C}}\}$ 
23:    FBP()
24:  end for
25:  for  $e \leftarrow E_3 + 1, \dots, E$  do # Phase 4
26:    FBP()
27:  end for
28:  Cache  $\mathcal{T}_t, \mathbf{m}_t, \bar{\mathbf{s}}_t$  for equation ??
29: end for
```
